

FULL OF SOUND AND FURY: The Recent History of Ancient DNA

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■ **Abstract** The discovery that DNA survives in ancient remains and can be amplified by the polymerase chain reaction has added a direct temporal dimension to evolutionary studies. Initial reports suggested that the time period open to investigation was vast, extending back into the Cretaceous period. However, attempts to replicate of results involving DNA purported to be over a million years old have not succeeded. Theoretical studies suggest that DNA is unlikely to survive intact more than about 100,000 years. However, even over this time period, the evolutionary questions that can be addressed are far reaching and include systematics, paleoecology, the origin of diseases, and evolutionary processes at the population level.

INTRODUCTION

“Would that I could discover truth as easily as I can uncover falsehood.”

Cicero, *De Natura Deorum* (44 BC)

The use of genetic data has greatly expanded evolutionary studies of extant organisms, with one major limitation: History must be reconstructed by extrapolation from current genetic patterns rather than directly observed in the fossil record. Systematists utilize character state data and phylogenetic methods to reconstruct evolutionary history, but this approach is fraught with problems caused by character reversals and parallelisms (134). Similarly, population genetics cannot deal well with the confounding effects of natural selection, genetic drift, gene flow, and population history without resorting to simplified models and assumptions (141, 151). Molecular evolutionary biologists have envied the paleontologist's unique access to historical information and longed to utilize the rich

information inherent in ancient DNA sequences. Such was the promise of research on DNA preserved in the remains of extinct organisms (19, 26, 101, 103, 135). However, the intense excitement at the first reports of ancient DNA greater than a million years old was followed by confusion and disillusionment when the claims were severely criticized (3, 58, 124, 135, 136, 155). As a result, the field has matured considerably, and while it remains thematically diverse, rigorous authentication procedures provide a unifying methodological focus. In this review, we attempt to reduce the confusion surrounding ancient DNA studies by first providing a historical overview and then focusing on studies that define directions for future research.

Ancient DNA: Definition and Scope

Ancient DNA deals exclusively with deoxyribonucleic acid isolated from plant or animal remains. It is therefore a subdiscipline of molecular paleontology, which more broadly concerns any biomolecule that can be extracted from ancient tissues such as pigments or polysaccharides from plant tissues (12, 62, 85, 118) or collagen and hemoglobin from animal remains (2, 18, 32, 86, 123). The time frame is broad for ancient DNA and includes the study of extant populations through the collection of hair, feces, or seeds (60, 73, 93), an endeavor often called “non-invasive sampling” (92). More commonly, however, ancient DNA has been used to describe a wide range of research efforts involving: (a) museum and archeological specimens (23, 25, 41, 48, 59, 61, 68, 100, 119, 142, 147); (b) subfossil remains from the late Pleistocene (51, 64, 71); and (c) fossils greater than a million years old (14, 29, 35, 110, 126).

The questions addressed by ancient DNA research concern any issue that benefits from a direct historical perspective. These areas include (a) systematics (23, 29, 36, 51, 61, 64, 67); (b) changes in genetic diversity as a function of time and environmental change (40, 52, 143); (c) migration and admixture (79, 91, 120); (d) ecology and paleoecology (73, 111); (e) the origin and spread of disease (8, 121); and (f) tempo and mode of mutations in populations (55). Consequently, the potential impact of ancient DNA studies is considerable; hence the enthusiasm and anticipation that greeted the first reports of DNA isolation from ancient remains (26, 103, 135).

THE HISTORY OF ANCIENT DNA

“Life’s but a walking shadow . . . It is a tale told by an idiot, full of sound and fury, signifying nothing.”

Shakespeare, *Macbeth* (1605–1606)

From the beginning, ancient DNA research was a populist science. Reports of DNA from ancient remains led to wild speculation in the press and film that life could be restored to ancient creatures. Each new discovery served to reconfirm the public

impression that scientists were moving quickly toward this goal. New reports of ancient DNA, although often of limited evolutionary significance, were published in the most prestigious journals. The first reports in 1985 were based on fairly recent material in museum vertebrate collections or human mummies and utilized cloning technology (59, 100). The initial report on mummies was spectacular because the length of DNA cloned was 3.4 kilobases (100). The preservation of such long fragments boded well for ancient DNA research; potentially entire genes could be resurrected. However, cloning ancient fragments requires a substantial source of DNA and is hit-or-miss with regard to specific single copy genes (101). In fact, the cloned sequences were uninformative, highly repetitive DNA later shown to be a part of the human HLA system and a probable contaminant (27).

Several years later, the advent of the polymerase chain reaction (PCR) permitted specific sequences to be amplified from only a few template molecules (94). As a result, the requirement for a substantial source of intact DNA was greatly reduced, and specific, phylogenetically informative genes could be targeted (61). For animals, the genes chosen were from the mitochondrial genome because there are thousands of copies in each cell. In addition, nearly a decade of research on genes such as *cytochrome b* and the hypervariable control region had demonstrated their utility for evolutionary questions ranging from population genetics to higher order systematics (5). For plants, the focus was on a chloroplast gene, 1,5 *bisphosphate carboxylase oxygenase* (*RbcL*), which had been sequenced in a wide variety of plant species owing to the availability of universal primers (22). In fact, the first DNA sequences from specimens more than a million years old were claimed for remarkably well-preserved 17–20-million-year-old magnolia leaf compression fossils from clay sediments of a Miocene shallow freshwater lake (35). Visible tissue structures and even cell walls and intracellular organelles could be seen in these ancient leaves. Phylogenetic analysis of supposed ancient sequences showed them to be most closely related to living *Magnolia* species, thus adding considerable weight to the claim of authenticity. Until the publication of the magnolia paper, ancient DNA sequencing was confined to specimens only a few thousand years old such as museum skins of the extinct Tasmanian wolf, *Thylacinus cynocephalus* (142), human mummies, and bog-preserved human brain tissue (30, 100). A year later, the presence of ancient DNA sequences in the magnolia leaves was apparently supported by an independent group using bald cypress specimens from the same deposit (126). However, the *Magnolia* study was performed without PCR controls (83), and other attempts to replicate these results have failed, raising strong doubts (105, 124, 127).

The magnolia study broke the million-year barrier, and with considerable excitement researchers turned to long extinct charismatic creatures such as dinosaurs as well as more mundane insects preserved in amber. A strong case could be made for amber preserved specimens because researchers believed that amber rapidly desiccates trapped remains and inhibits bacterial activity, two prerequisites for preservation of ancient DNA (14, 109). The list of amber preserved specimens from which ancient sequences were recovered is impressive and includes stingless

bees, termites, honeybees, and plants (14–16, 29, 110). Even fossil bacterial sequences were claimed from amber (13), and recovered spores were viably cultured (17), although these sequences were subsequently shown to be probable contaminants (154). Additionally, the presence in extracts of contaminating sequences from other extant insect species was worrisome (28, 29), although the practice of reporting such contaminants was not widespread. More recent studies on chemical preservation of plant and animal remains in Oligocene amber show DNA is unlikely to survive (129).

The grand culmination of studies using specimens in the million-year-plus range was the report of dinosaur mtDNA sequences from an unlikely source: partially carbonized material from a coal deposit in Idaho (150). The sequences appeared neither mammalian nor reptilian and were therefore assumed to be endogenous. Science fiction was finally fact. However, although the methods and results seemed reasonable, given past research, existing authentication standards were not met (83), and the dinosaur DNA was soon shown by several groups to be an artifact (1, 56, 58, 153, 155). Comprehensive phylogenetic analysis showed that the supposed dinosaur sequences were classified with mammals and were closest to humans. An elegant study from the laboratory of Svante Pääbo, the leading critic and practitioner of ancient DNA research, showed that the putative dinosaur DNA was probably a previously undescribed human intron (156). A similar fate befell the study of DNA from Cretaceous dinosaur eggs (146).

Pääbo and colleagues have also criticized ancient DNA research on other grounds such as the lack of suitable controls, facilities, and techniques to limit contamination (102, 103). The power of PCR was and is the problem: A single intact contaminating sequence from a recent source can potentially outcompete ancient degraded and damaged DNA in the process of amplification. Remarkably, recent contaminating DNA could be mixed into ancient sequences by “jumping PCR” (103). Chimeric molecules are formed when damaged ancient DNA templates are incompletely copied during a PCR cycle, allowing the incomplete extension product itself to act as a primer in the next round of PCR. Damaged DNA can cause many problems during PCR including erroneous insertion of adenosine residues (104) and inaccurate amplification of repetitive motifs as well as the production of chimeric molecules from endogenous and contaminating sequences (28). On phylogenetic grounds, chimeric sequences could appear quite reasonable and thus could be mistaken as authentic. The sources of contamination are numerous and unpredictable, ranging from those at the point of excavation to curatorial activities and laboratory reagents and aerosols (48).

Yet another problem is “PCR carry-over,” which occurs when PCR products from one reaction contaminate those of another (78). The scale of this problem can be considerable. For example, a single 100 μ l PCR reaction produces enough template molecules that, if diluted in an Olympic size swimming pool, each 100 μ l from the pool would still contain 400 intact copies (78). The possibility, therefore, of contaminants from clothes, reagents, and the atmosphere in genetic laboratories is immense. This problem is less severe in studies of nonhumans because a

laboratory can be chosen in which the target taxon and related species have never been studied. However, even animal studies are not immune from contamination; despite careful controls, sequences related to *Drosophila* have been reported in extracts of amber termites (29).

Contamination problems take on a new dimension when human remains are used as a source of ancient DNA (24, 102, 133). However, numerous researchers have claimed success with ancient human specimens hundreds to thousands of years old (97). The early reports simply documented the presence of endogenous sequences (30, 80, 99, 100, 149), whereas more recent studies have addressed questions of kinship, sex ratios, disease, and large-scale patterns of human colonization (8, 41, 44, 46, 90). Notably, studies on ancient Americans imply that they had a different genetic composition than do present-day native Americans, thus casting doubt on colonization theories based on recent samples alone (79, 90, 91, 107, 122, 132). The pattern of settlement of Polynesia also is controversial. It appears to have been settled in several waves, contradicting the "fast track" theory of settlement from Asia rather than Melanesia; however, no consensus has been reached (42, 44). Few of these studies have been independently replicated. Importantly, DNA analysis of ancient remains from areas separated by only a few hundred kilometers do not agree (30, 107). These discrepancies have been variously explained as sampling effects or evidence of local spatial or temporal differentiation (97).

The possibility of recent human DNA contamination and the likelihood of jumping PCR cast doubt on the verity of the human results (48). Some sequences or parts thereof may be authentic, but it is difficult to assess without independent replication how much is truth or falsehood. Very troubling is the absence of success of ancient human studies in the Pääbo laboratory where conditions were carefully controlled for contamination: 110 ancient Egyptian bone and tissue samples and 45 ancient remains from the New World yielded little amplifiable DNA (49, 65). More convincing are cases where some genetic information is known about the specimens in advance. For example, bones sexed by morphologic methods had the expected sex chromosome markers, strongly suggesting the DNA was endogenous (131). No simple and elegant solution has emerged for the problems of contamination (3, 4), although independent replication is essential. The minimum requirements for research with ancient human remains are to extract DNA from ancient samples in a separate facility with air filtration, UV sterilization, and protective sterile clothing and to use a carefully considered set of negative and positive controls. Even then, contaminants are common (50, 76) or sometimes surprisingly absent (41, 63, 79).

Cloning and sequencing of PCR products have revealed recombinants and allowed ancient sequences to be reconstructed by separating the contaminants from endogenous sequences (28). However, this approach assumes that the contaminating partial sequences can be identified. Importantly, recombinant sequences should be highly context dependent; thus, a separate amplification from the original extract or an independent laboratory should not obtain the same recombinant sequences. Therefore, a first proof of authenticity is replication. Sadly, no million-year-old

DNA has been replicated in this fashion, despite considerable effort (4, 145). Since 1995, no reliable reports of DNA past the million-year mark have been published. Given theoretical and empirical results, the view now is that specimens much older than 100,000 years in age are unlikely to contain endogenous DNA that can be amplified (82, 84).

The gold standard of authentication in studies of ancient DNA has been set by Pääbo and his collaborators (49, 50, 76, 77). Their general experimental design is exemplified by DNA analysis of the "Tyrolean Ice Man," a 5000-year-old human discovered frozen in the Italian Alps (50). DNA was extracted in a special facility for ancient DNA with extraction controls; results were replicated independently, and PCR products were cloned and screened for contaminants. However, the theoretical importance of the study for human evolution was negligible; their considerable effort showed only that the iceman was European rather than a South American mummy hoax. More consequential was the carefully documented DNA analysis of the Neanderthal-type specimen (76, 77). Here, PCR products were originally cloned and sequenced by the Pääbo laboratory, and sequences were examined for persistent motifs that differed from those in living humans. These sequence differences were then used to design Neanderthal-specific PCR primers that would not amplify contaminating human DNA. Using these specific primers, the resulting Neanderthal mitochondrial control region sequence was replicated in an independent laboratory and was shown to be phylogenetically divergent from recent humans. Consequently, the results imply that the Neanderthals were a separate hominid lineage that did not significantly contribute genetically to modern humans.

THE REAL ANCIENT DNA

Theoretical and empirical studies have demonstrated that DNA is best preserved in cold and dry environments protected from UV exposure and in remains less than 100,000 years old (82, 84, 101, 112). DNA has been recovered from animal remains preserved in the past 50,000 years from the Arctic permafrost by several groups (47, 65, 96, 98, 152), confirming that DNA is well preserved under these conditions. Other environments conducive to preservation are those in high altitude cold, dry caves (40, 64). DNA has also been reported from ancient human remains only a few thousand years old under a variety of conditions, including exposure to water where hydrolytic damage to DNA might be expected (30, 55). However, as discussed above, considerable uncertainty exists about the actual presence of endogenous DNA sequences. The more convincing studies are those in which DNA sequences match that expected from independent species identification of archeological material (21). For example, pig and sheep bones recovered from the *Mary Rose*, a British ship sunk in the sixteenth century, provided the expected sequences for these species as well as those of human contaminants (43).

Despite the limitations imposed by time and preservation conditions, some deposits hold great potential for the preservation of DNA. Most promising are

remains preserved in the Arctic permafrost. Hundreds of thousands of bones recovered from permafrost deposits are housed in museum collections in North America and Siberia, representing large numbers of mammal and bird species. Recent research has focused on the systematics of the largest of the permafrost species, the mammoth and mastodons of Alaska and Siberia. Mammoths appear most closely related to recent elephants, as expected, although the topology of this clade is still unresolved (47, 67, 98, 152). Other species such as horses and woolly rhinos also have provided DNA sequences (9, 65). The hope is that with further sampling, population-level questions can be resolved as well. Most permafrost samples range in age from about 10,000 to 50,000 years before present (38), and hence they provide a genetic record of diversity before and after the last glaciation. Population declines associated with the last glaciation and the Late Pleistocene large mammal extinctions (37, 39) are likely to have had significant effects on genetic diversity. Ancient DNA sequences from a dated series of specimens would provide a direct record of these genetic changes and of faunal interchanges between the Old and New World across the Bering land bridge.

The most spectacular record of Late Pleistocene carnivores is without question that preserved in the Rancho La Brea tar pits at the George C. Page Museum in Los Angeles. Several thousand specimens of carnivores and, to a much lesser extent, large ungulates, are housed in the museum collection and span a time period from about 8,000 to 40,000 years before present. DNA was reported from marrow bone samples of a saber-toothed cat, *Smilodon fatalis*, after removal of embedded tar using kerosene (71). DNA segments from mitochondrial 12S *rRNA* and the nuclear *FLA-I* gene were amplified and shown to be cat-like. Phylogenetic analysis of the sequences appeared to group the saber-tooth cats closely with modern cats rather than paleofelids (Nimravidae). However, the sequence data did not provide enough resolution to definitively align the saber-tooth with a specific group of living cats and was not replicated. Consequently, the possibility of large-scale analysis of tar pit fossils to address systematic or population level questions remains an interesting possibility.

DNA sequences have been amplified from a diversity of species preserved in caves at a variety of locations. An early and dramatic report concerned the isolation and amplification of mitochondrial DNA sequences from five species of extinct New Zealand moa (Dinornithiformes) recovered principally from caves (23). New Zealand has been separated from Australia for about 80 million years, and it was therefore assumed that these flightless birds shared a common ancestor with the other endemic ratite, the kiwi. However, phylogenetic analysis of 12S RNA sequences showed that the moa represents an early offshoot of the ratites, whereas the kiwi was closely related to the Australian emu and cassowary. Consequently the data indicate that ratite birds had independently colonized New Zealand twice. DNA from teeth of pocket gophers (40), cave bears (51), rabbits (54) and ground sloths (64) are other examples of important studies based on cave material.

Archeological material preserved by burial in temperate soils has provided a source of DNA that can be amplified by PCR (87). The European rabbit,

Oryctolagus cuniculus, has been best studied in this regard. Mitochondrial DNA has been amplified from 90 rabbit bones from 22 localities, ranging over the past 11,000 years (52). Additionally, an island population off the Tunisian coast that was founded over 1400 years ago was studied (53). The research showed genetic continuity over much of the recent history of the European rabbit, but during the Middle Ages, a new group of mitochondrial *cytochrome b* genotypes appeared in France that are ancestral to modern domestic rabbits. This result suggests that rabbits from this lineage were favored and may reflect the establishment of warrens in the tenth century. Finally, insight into the process of domestication was gained through the study of ancient cattle, *Bos primigenius*, and their domestic descendants *B. taurus* (6). Sequences from remains of primitive cattle as old as 12,000 years before present were directly ancestral to modern European breeds. This further supports genetic analyses of recent cattle that suggested that European cattle were domesticated from a different subspecies than were Asian breeds.

THE FUTURE OF ANCIENT DNA RESEARCH

The number of ancient DNA papers has declined in the past few years as has the number of reports appearing in high profile journals. In the first decade of ancient DNA research, 35% of articles appeared in *Nature*, *Science*, or the *Proceedings of the National Academy of Science of the USA*, but in the last three years this proportion has declined to less than 15%. The more recent proportion is similar to that for 300 randomly chosen papers in evolutionary biology (7.6%). The honeymoon period has passed for ancient DNA research, and the difficulties associated with a maturing field need confronting. The absence of replication of million-year-old DNA studies and the theoretical and empirical research on DNA preservation all suggest that DNA is unlikely to be preserved much beyond 100,000 years (82, 112). Therefore, only a small fraction of time can be traced through the study of ancient DNA. Nevertheless, as outlined below, we feel that several research topics can dependably be addressed by studies of ancient DNA from specimens younger than 100,000 years.

Systematics of Recently Extinct Species

The most common use of ancient DNA has been directed at the systematics of extinct species and was first demonstrated by a study of the extinct Tasmanian wolf (*Thylacinus cynocephalus*), a marsupial analog of the gray wolf that vanished in the early part of the twentieth century (142). *Cytochrome b* sequences demonstrated that the species was closely related to other Australian marsupials rather than to South American carnivorous marsupials as some had supposed. More extensive analysis has confirmed this initial result (75). Similarly, *cytochrome b* sequences amplified from museum specimens of the extinct quagga (*Equus quagga*) clearly

demonstrated its relationship to the mountain zebra (*E. zebra*) rather than to the horse (*E. caballus*) (59, 61).

Cave deposits have provided important systematic insights for mammals as well. A control region sequence from a European late Pleistocene cave bear (*Ursus spelaeus*) sample was shown to be from a distinct species with a sequence basal to those found in Old and New World populations of brown bears, *Ursus arctos* (51). However, cave material often may not contain DNA that can be amplified; only two of 35 samples of the ground sloth *Myiodon darwini* from a variety of settings in North and South America provided DNA that could be amplified (64). The two successful 13,000-year-old samples were from Myiodon cave, a cold, dry cave in southern Chile. Phylogenetic analysis of mitochondrial genes encoding for 12S and 16S rRNA clearly showed that the ground sloth was unexpectedly sister to two-toed rather than three-toed sloths, implying that arboreality in sloths evolved twice. This type of study will require more effort in the future, as there are severe limitations on phylogenetic analysis using very short stretches of DNA. Longer sequences, composed of multiple overlapping fragments, will be necessary to improve the resolution and confidence of phylogenies.

Genetic Diversity Over Time

One of the earliest, underappreciated reports on ancient DNA concerned a study of temporal continuity in kangaroo rats (*Dipodomys panamintinus*) collected in 1911, 1917, and 1937 from three areas in central California and representing three different subspecies (143). Researchers sampled the three localities in 1988, after some 25 to 50 generations had elapsed. This is sufficient time for dominance of a mitochondrial control region matriline to change through drift or through recolonization after an extinction event. However, in each population, the genetic diversity appeared unchanged, and the distribution of control region genotypes was not significantly different from that expected in a population with high levels of genetic diversity.

This research set the stage for other efforts where temporal changes were explicitly expected. Endangered species are a model group in this regard because many have had their population sizes reduced dramatically within the past hundred years, and their declines have been periodically sampled by museum collectors. In fact, some problematic carnivore species that disappeared from the United States such as the gray wolf (*Canis lupus*) and eastern puma (*Felis concolor*) were exterminated by government officials who incidentally sent some specimens to museums. The red wolf (*Canis rufus*) is an example. Analysis of early twentieth century skins showed that mitochondrial *cytochrome b* sequences were not distinct from those in gray wolves (*C. lupus*) or coyotes (*Canis latrans*), suggesting that the red wolf was a hybrid species (147). Later, more extensive research on nuclear and other mitochondrial genes confirmed the early report (120).

By comparison with more abundant related species, endangered populations are often found to be low in genetic variability (34). However, differences in life

history or low predisturbance abundance may be the reason for current lower levels of diversity. Hence a direct temporal perspective is preferred. A study of the endangered Australian northern hairy-nosed wombat (*Lasiorninus krefftii*) demonstrated the problems with the comparative approach (140). Here, samples from the last existing wombat colony of about 50 individuals had low levels of variation relative to a related species. However, the levels of variation were similar to that of an extinct pre-bottleneck northern hairy-nosed wombat population represented by museum skins. Thus, low levels of variation may have been characteristic of northern hairy-nosed wombats before the decline. In contrast, historic feather samples from the endangered San Clemente Island loggerhead shrike (*Lanius ludovicianus mearnsi*) and greater prairie chicken (*Tympanuchus cupido*) demonstrated genetic continuity but marked reductions in genetic variation in the extant populations (10, 95).

Reconstructing the past population relationships of endangered species also is important for reintroduction programs. In the wombat study, the past geographic range was uncertain because partial skeletal remains were not definitive. Mitochondrial DNA sequencing clearly established that a far southern population once existed. Similarly, subfossil bones of the Laysan duck (*Anas laysanensis*) were identified by mitochondrial DNA analysis from lava tubes on the main Hawaiian islands where they apparently had gone extinct (25). These data justified reintroduction and suggested that many island endemics may be relics of former cosmopolitan species (130). Museum collections have become a source of DNA from populations that are difficult to sample or are extinct, so much so that most museums now have a specific loan policy and review board for destructive DNA sampling (20, 148).

Migration and Admixture

Some fossil deposits hold the potential for temporal analysis of genetic changes at the population level. For example, owl pellet deposits in caves provide a temporal sampling of rodent populations and preserve DNA well (138). Owls roosting near cave openings and depositing pellets create a rich stratigraphic collection of their prey. The high altitude Lamar Cave in Yellowstone National Park was excavated and yielded several thousand pocket gopher (*Thomomys talpoides*) mandibles from 16 dated stratigraphic levels (40). *Cytochrome b* sequences were obtained from 73 of 88 teeth representing these stratigraphic levels and showed that genetic continuity characterized the three-thousand-year history of the deposit. The population near Lamar Cave was apparently isolated from genetic exchange from populations only a few hundred kilometers distant. In contrast, phenotypic changes occurred in this population that reflected changing climates. Such phenotypic change must be due to within-population processes and does not involve the differential migration of individuals having morphologic attributes more appropriate for the altered environment. The result has important conservation implications (40, 81); it implies that in small nonvagile species, phenotypic changes may occur in response to climate change but are limited by the existing genetic variation within populations.

Ecology and Paleoecology

DNA sequences can often be retrieved from remains deposited by organisms as part of their life cycle. For vertebrates, this might include hair, feathers, scales, feces, saliva, and urine (92). For invertebrates, pupal casts, eggs, and sperm could provide a source of DNA as well as museum specimens. A variety of tissues are discarded by plants; DNA sequences have been retrieved from herbarium samples of leaves, flowers, and seeds (128). Noninvasive sampling is often the only means to do genetic analysis in endangered species and has been used to understand population relationships and diversity in the brown bear (*Ursus arctos*) from hair samples left on scrapes (137). However, feces provide a more comprehensive source of information as they are often left in predictable areas and contain a genetic record not only of the DNA of an individual but also that of ingested foods and internal microorganisms (66, 73, 114).

The spatial distribution of feces, identified by genetic techniques to an individual, provides information about population density, home range, and movement patterns. A systematic collection of coyote (*Canis latrans*) feces from a southern California parkland permitted investigators to more accurately estimate population size, sex ratio, and movement patterns that were not apparent from radiotelemetry studies (74). Fecal analysis is not limited to modern samples. Fecal DNA was elegantly exploited in a study on 11,000-year-old ground sloth dung (111). Here aspects of diet and altitudinal movement patterns could be inferred from DNA analysis of feces. Considering that coprolites of large mammals are not uncommon, they may provide a unique window into paleoecology of extinct Late Pleistocene mammals. Other studies have used hair samples from chimpanzee (*Pan troglodytes*) night nests to deduce paternity and mating patterns (93), and samples of whale meat sold in Japanese markets have identified endangered specimens banned under international convention (7, 106). Forensic applications of ancient DNA analysis are numerous (33, 45, 72).

Disease

Disease outbreaks often begin with a few cases over a limited geographic area. However, the pathogenic organisms causing the disease may have a wider distribution for some time previous to the first reported cases. DNA analysis of museum or archeological specimens can provide a more exact temporal and spatial perspective on the initiation and spread of disease. For example, analysis of ticks from deer skins in museum collections showed that Lyme disease (*Borrelia burgdorferi*) was present nearly a decade earlier than the first reported cases in the United States and Europe (88, 108). Sequences from *Mycobacterium tuberculosis*, *M. leprae*, *Yersinia pestis*, and the 1918 "Spanish" influenza have variously been reported from samples of ancient bone, and teeth, medical samples, and permafrost-preserved soft tissue (8, 11, 31, 113, 115, 121, 139), suggesting a potential field of genetic paleopathology. Unfortunately, some pathogens are common

in the extant human population (e.g. *M. tuberculosis*) or have congeneric relatives common in soil or other animal hosts (e.g. *Mycobacterium bovae*), raising the possibility of contamination through handling, burial, or the use of animal products such as BSA. The possibility of identifying genetic disease in historical figures such as the Marfan syndrome in Abraham Lincoln has been discussed (89). Disease greatly impacts the dynamics of wild populations, and much can be learned from understanding the geographic origin of pathogenic organisms (116). Disease organisms that leave a DNA record in preserved tissue can potentially be detected if the analysis is done with appropriate controls and in laboratories that have had no history of research on the disease-causing organism.

Mutational Dynamics

Some populations are isolated for evolutionary time periods. Therefore, mutations that appear in these populations provide a direct estimate of mutation and fixation rates. Considerable controversy exists on the mutation rate of mitochondrial genes as there is an order of magnitude difference in mutation rates from germline studies and those based on evolutionary divergence (69). Dependence between observed rates and time is implied. As both mutation rate estimates involve extrapolation, a direct measure of mutation rates in an isolated population afforded by sampling over time would be more desirable. A few studies have hinted at this possibility. For example, a rate of control region evolution was estimated from mutations appearing in 2000 years of human remains at a single site (55) and from remains of primitive cattle (6). In both cases, the estimated rate was similar to that predicted by indirect phylogenetic methods. However, neither population was likely to be geographically isolated, and thus gene flow remains a source of new alleles. Studies of isolated populations in which the temporal record is good are needed. For example, the Great Salt Lake deposits of Utah are dominated by cysts from brine shrimp, *Artemia*, that may contain preserved DNA as well (MT Clegg, personal communication). Genetic analysis of these shrimp over thousands of years is conceivable with large population samples of each time interval.

PERSPECTIVES AND CONCLUSIONS

The rise and decline of ancient DNA research has been dependent on the development of new techniques, the limits of which subsequently were defined by critical analysis and replication. The development of new techniques and methods of analysis still promises new avenues of research (e.g. sloth dung), but the realm of questions is now circumscribed by the likelihood that DNA will not persist much beyond 100,000 years. Given this reality, efforts need to focus on the application of existing technology to consequential issues that can be resolved over this time scale. Clearly, systematics of recently extinct species, some of which

may have deep phylogenetic roots such as the giant moa and the extinct ground sloth, help resolve larger evolutionary issues. Additionally, systematics of recently extinct species potentially provides an outgroup perspective that can greatly improve confidence in topologies of phylogenetic trees of extant taxa (29, 76, 98, 152).

The genetic record of temporal changes in populations is perhaps the most intriguing and least explored potential of ancient DNA research over millennia time scales. Prior to the discovery of ancient DNA, genetic changes within populations were reconstructed from current patterns of genetic diversity. This static view often assumed equilibrium conditions or specific demographic models of population growth and expansion (70, 117, 151). Given the dynamics of climate change and migration between and within continents, such assumptions are likely to be violated. Ancient DNA analysis can provide a direct record of the tempo and mode of genetic change within populations, and thus a means of testing and refining existing population models. The deposits with the most potential are those in Arctic permafrost, high altitude caves, and other cold, arid environments. In combination with stratigraphic information and radiocarbon dating, analysis of these remains provides an absolute temporal scale to view the mutational and demographic processes occurring within and among populations.

Conservation biology may be well served by ancient DNA analysis. Systematics of extinct species provides an important perspective on the remaining diversity of extant organisms. For example, research on avian remains from Polynesian islands suggests that over 2000 species (one quarter of all bird species) may have gone extinct in historic times (130). Molecular systematics of these extinct species would place their extinction in a phylogenetic context and better highlight the evolutionary diversity that was lost (144). Additionally, insights into island speciation and colonization could result. Population level analysis also has important conservation implications. Analysis of genetic diversity of past populations of endangered species establishes a baseline from which the loss of genetic variability can be measured. Additionally, geographic sampling from museum or archeological collections allows reconstruction of the past geographic limits of endangered species. These limits, in combination with the inferred genetic relationships among populations, provide a guide for reintroduction and augmentation (57). Genetic changes that accompany climatic or environmental shifts may also be uncovered by ancient DNA analysis. For example, the body size of the bushy-tailed woodrat (*Neotoma cinerea*), as estimated by the diameter of their feces, dramatically decreased since the last glaciation in association with global warming (125). DNA analysis of these feces may be possible with new techniques (111) and would allow the dissection of the causes of phenotypic change; do they reflect changes in species composition or evolutionary responses within populations?

Finally, DNA analysis of fecal remains and other tissue deposited by organisms may provide information about diet, movements, and past pathogens. Although some may despair over the unlikely prospect of Cretaceous DNA, most will

recognize that the study of ancient DNA, if directed toward millennial scale changes, still has the potential to solve evolutionary mysteries. The initial excitement may be gone from the field, but with maturity of purpose, it will come to signify more than sound and fury.

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LITERATURE CITED

- Allard MW, Young D, Huyen Y. 1995. Detecting dinosaur DNA. *Science* 268:1192–93
- Ambler R, Daniel M. 1991. Proteins and molecular palaeontology. *Philos. Trans. R. Soc. London Ser. B* 333:381–89
- Audic S, Béraud-Colomb E. 1997. Ancient DNA is 13 years old. *Nat. Biotechnol.* 15:855–58
- Austin JJ, Ross AJ, Smith AB, Fortey RA, Thomas RH. 1997. Problems of reproducibility—does geologically ancient DNA survive in amber-preserved insects? *Proc. R. Soc. London Ser. B* 264:467–74
- Avise J. 1994. *Molecular Markers, Natural History, and Evolution*. New York: Chapman & Hall
- Bailey J, Richards MB, Macaulay VA, Colson IB, James IT, et al. 1996. Ancient DNA suggests a recent expansion of European cattle from diverse wild progenitor species. *Proc. R. Soc. London Ser. B* 263:1467–73
- Baker CS, Cipriano F, Palumbi SR. 1996. Molecular genetic identification of whale and dolphin products from commercial markets in Korea and Japan. *Mol. Ecol.* 5:671–85
- Baron H, Hummel S, Herrmann B. 1996. *Mycobacterium tuberculosis* complex DNA in ancient human bones. *J. Archaeol. Sci.* 23:667–71
- Best CH. 1994. Genetic analysis of ancient DNA from the hair of the woolly rhinoceros *Coelodonta antiquitatis*. *Proc. Arctic Sci. Conf., 45th, Anchorage*, pp. 37. Vladivostok: Dalnauka
- Bouzat JL, Lewin HA, Paige KN. 1997. The ghost of genetic diversity past resurrected: ancient DNA analysis of the greater prairie chicken *Tympanuchus cupido*. *Bull. Ecol. Soc. Am.* 78:57
- Braun M, Cook D, Pfeiffer S. 1998. DNA from *Mycobacterium tuberculosis* complex identified in north American, pre-Columbian human skeletal remains. *J. Archaeol. Sci.* 25:271–77
- Brown T, Allaby R, Brown K, Jones M. 1993. Biomolecular archaeology of wheat: past, present and future. *World Archaeol.* 25:64–73
- Cano R, Borucki MK, Higby-Shweitzer M, Poinar HN, Poinar GO, Pollard KJ. 1994. *Bacillus* DNA in fossil bees: an ancient symbiosis? *Appl. Environ. Microbiol.* 60:2164–67
- Cano R, Poinar H, Pieniazek N, Acra A, Poinar G. 1993. Amplification and sequencing of DNA from a 120–135 million year old weevil. *Nature* 363:536–38

15. Cano R, Poinar H, Poinar G. 1992. Isolation and partial characterization of DNA from the bee *Proplebeia dominicana* (Apidae: Hymenoptera) in 25–40 million year old amber. *Med. Sci. Res.* 20:249–51
16. Cano R, Poinar H, Roubik D, Poinar G. 1992. Enzymatic amplification and nucleotide sequencing of portions of the 18S rRNA gene of the bee *Proplebeia dominicana* isolated from 25–40 million year old amber. *Med. Sci. Res.* 20:619–22
17. Cano RJ, Borucki M. 1995. Revival and identification of bacterial spores in 25- to 40-million-year-old Dominican amber. *Science* 268:1060–64
18. Cattaneo C, Gelstorp K, Phillips P, Sokol R. 1995. Differential survival of albumin in ancient bone. *J. Archaeol. Sci.* 22:271–76
19. Cherafas J. 1991. Ancient DNA: still busy after death. *Science* 253:1354–56
20. Cherry MI. 1994. Ancient DNA and museums. *S. African J. Sci.* 90:437–38
21. Cipollaro M, Di Bernardo G, Galano G, Galderisi U, Guarino F, et al. 1998. Ancient DNA in human bone remains from Pompeii archaeological site. *Biochem. Biophys. Res. Commun.* 247:901–4
22. Clegg MT, Zurawski G. 1992. Chloroplast DNA and the study of plant phylogeny: present status and future prospects. In *Molecular Systematics of Plants*, ed. PS Soltis, JE Soltis, JJ Doyle, pp. 1–13. New York: Chapman & Hall
23. Cooper A, Mourer-Chauviré C, Chambers GK, von Haeseler A, Wilson A, Pääbo S. 1992. Independent origins of New Zealand moas and kiwis. *Proc. Natl. Acad. Sci. USA* 89:8741–44
24. Cooper A. 1997. Reply to Stoneking: ancient DNA—How do you really know when you have it? *Am. J. Hum. Genet.* 60:1001–2
25. Cooper A, Rhymer J, James HF, Olson SL, McIntosh CE, et al. 1996. Ancient DNA and island endemics. *Nature* 381:484
26. Curry G. 1987. Molecular palaeontology: new life for old molecules. *Trends Ecol. Evol.* 2:161–65
27. Del Pozzo G, Guardiola J. 1989. Mummy DNA fragment identified. *Nature* 339:431–32
28. DeSalle R, Barcia M, Wray C. 1993. PCR jumping in clones of 30 million year old DNA fragments from amber preserved termites (*Mastotermes electrodominicus*). *Experientia* 49:906–9
29. DeSalle R, Gatsley G, Wheeler W, Grimaldi D. 1992. DNA sequences from a fossil termite in Oligo-Miocene amber and their phylogenetic implications. *Science* 257:1933–36
30. Doran GH, Dicke DN, Ballinger WE Jr, Agee OF, Laipis PJ, Hauswirth WW. 1986. Anatomical, cellular and molecular analysis of 8000-yr-old human brain tissue from the Windover archaeological site. *Nature* 323:803–6
31. Drancourt M, Aboudharam G, Signoli M, Dutour O, Raoult D. 1998. Detection of 400-year-old *Yersinia pestis* in human dental pulp: an approach to the diagnosis of ancient septicemia. *Proc. Natl. Acad. Sci. USA* 95:12637–40
32. Evershed R. 1993. Biomolecular archaeology and lipids. *World Archaeol.* 25:74–93
33. Fisher D, Holland M, Mitchell L, Sledzik P, Wilcox A, et al. 1993. Extraction, evaluation, and amplification of DNA from decalcified and uncalcified United States civil war bone. *J. Forensic Sci.* 38:60–68
34. Frankham R. 1996. Relationship of genetic variation to population size in wildlife. *Conserv. Biol.* 10:1500–8
35. Golenberg E, Giannasi D, Clegg M, Smiley C, Durbin M, et al. 1990. Chloroplast sequence from a Miocene *Magnolia* species. *Nature* 344:656–58
36. Goloubinoff P, Pääbo S, Wilson A. 1993. Evolution of maize inferred from sequence diversity of an *adh2* gene segment from archaeological specimens. *Proc. Natl. Acad. Sci. USA* 90:1997–2001

37. Graham RW, Lundelius EL Jr, Graham MA, Schroeder EK, Toomey RS III, et al. 1996. Spatial response of mammals to late Quaternary environmental fluctuations. *Science* 272:1601–6
38. Guthrie RD. 1990. *Frozen Fauna of the Mammoth Steppe: The Story of Blue Babe*. Chicago: Univ. Chicago Press
39. Guthrie RD. 1995. Mammalian evolution in response to the Pleistocene-Holocene transition and the break-up of the mammoth steppe: two case studies. *Acta Zool. Cracoviensis* 38:139–54
40. Hadly E, Kohn M, Leonard J, Wayne R. 1998. A genetic record of population isolation in pocket gophers during Holocene climate change. *Proc. Natl. Acad. Sci. USA* 95:6893–96
41. Hagelberg E. 1995. Mitochondrial DNA polymorphisms in modern and ancient Pacific Islanders. *Am. J. Phys. Anthropol.* 20:103
42. Hagelberg E. 1997. Ancient and modern mitochondrial DNA sequences and the colonization of the Pacific. *Electrophoresis* 18:1529–53
43. Hagelberg E, Clegg J. 1991. Isolation and characterization of DNA from archaeological bone. *Proc. R. Soc. London Ser. B* 244:45–50
44. Hagelberg E, Clegg J. 1993. Genetic polymorphisms in prehistoric Pacific islanders determined by analysis of ancient bone DNA. *Proc. R. Soc. London Ser. B* 252:163–70
45. Hagelberg E, Gray I, Jeffreys A. 1991. Identification of the skeletal remains of a murder victim by DNA analysis. *Nature* 352:427–29
46. Hagelberg E, Quevedo S, Turbón D, Clegg JB. 1994. DNA from ancient Easter islanders. *Nature* 369:25–26
47. Hagelberg E, Thomas MG, Cook CEJ, Sher AV, Baryshnikov GF, Lister AM. 1994. DNA from ancient mammoth bone. *Nature* 370:333–34
48. Handt O, Höss M, Krings M, Pääbo S. 1994. Ancient DNA: methodological challenges. *Experientia* 50:524–29
49. Handt O, Krings M, Ward RH, Pääbo S. 1996. The retrieval of ancient human DNA sequences. *Am. J. Hum. Genet.* 59:368–76
50. Handt O, Richards M, Trommsdorff M, Kilger C, Simanainen J, et al. 1994. Molecular genetic analysis of the Tyrolean Ice Man. *Science* 264:1775–78
51. Hänni C, Laudet V, Stehelin D, Taberlet P. 1994. Tracking the origins of the cave bear (*Ursus spelaeus*) by mitochondrial DNA sequencing. *Proc. Natl. Acad. Sci. USA* 91:12336–40
52. Hardy C, Callou C, Vigne J-D, Casane D, Dennebouy N, et al. 1995. Rabbit mitochondrial DNA diversity from prehistoric to modern times. *J. Mol. Evol.* 40:227–37
53. Hardy C, Casane D, Vigne JD, Callou C, Dennebouy N, et al. 1994. Ancient DNA from bronze age bones of European rabbit (*Oryctolagus cuniculus*). *Experientia* 22:487–89
54. Hardy C, Vigne J, Casan D, Dennebouy N, Mounolou J, et al. 1994. Origin of European rabbit (*Oryctolagus cuniculus*) in a Mediterranean island: zooarchaeology and ancient DNA examination. *J. Evol. Biol.* 7:217–26
55. Hauswirth WW, Dickel CD, Rowold DJ, Hauswirth MA. 1994. Inter-and intra population studies of ancient humans. *Experientia* 50:585–91
56. Hedges SB, Schweitzer MH. 1995. Detecting dinosaur DNA. *Science* 268:1191–92
57. Hedrick PW. 1995. Gene flow and genetic restoration: the Florida panther as a case study. *Conserv. Biol.* 9:996–1007
58. Henikoff S. 1995. Detecting dinosaur DNA. *Science* 268:1192
59. Higuchi R, Bowman B, Freiberger M, Ryder O, Wilson A. 1984. DNA sequence from the quagga, an extinct member of the horse family. *Nature* 312:282–84
60. Higuchi R, von Beroldingen C, Sensabagh

- G, Erlich H. 1988. DNA typing from single hairs. *Nature* 332:543–46
61. Higuchi R, Wruck L, Oakes E, George M, Tong B, et al. 1987. Mitochondrial DNA of the extinct quagga: relatedness and extent of postmortem change. *J. Mol. Evol.* 25:283–87
 62. Hillman G, Wales S, McLaren F, Evans J, Butler A. 1993. Identifying problematic remains of ancient plant foods: a comparison of the role of chemical, histological and morphological criteria. *World Archaeol.* 25:94–121
 63. Horai S, Kondo R, Murayama K, Hayashi S. 1991. Mitochondrial DNA sequence in contemporary and ancient humans. *Am. J. Hum. Genet.* 49:561
 64. Höss M, Dilling A, Currant A, Pääbo S. 1996. Molecular phylogeny of the extinct ground sloth *Myodon darwini*. *Proc. Natl. Acad. Sci. USA* 93:181–85
 65. Höss M, Jaruga P, Zastawny T, Dizdaroğlu M, Pääbo S. 1996. DNA damage and DNA sequence retrieval from ancient tissues. *Nucleic Acids Res.* 24:1304–7
 66. Höss M, Kohn M, Pääbo S, Knauser S, Schroder W. 1992. Excremental analysis by PCR. *Nature* 359:199
 67. Höss M, Pääbo S, Vereshchagin N. 1994. Mammoth DNA sequences. *Nature* 370:333
 68. Houde P, Braun M. 1988. Museum collections are a source of DNA for studies of avian phylogeny. *Auk* 105:773–76
 69. Howell N, Kubacka I, Mackey DA. 1996. How rapidly does the human mitochondrial genome evolve? *Am. J. Hum. Genet.* 59:501–9
 70. Hudson RR. 1990. Gene genealogies and the coalescent process. *Oxford Surv. Evol. Biol.* 7:1–44
 71. Janczewski D, Yukhi N, Gilbert D, Jefferson G, O'Brien S. 1992. Molecular phylogenetic inference from saber-toothed cat fossils of Rancho La Brea. *Proc. Natl. Acad. Sci. USA* 89:9769–73
 72. Jeffreys A, Allen M, Hagelberg E, Sonnerberg A. 1992. Identification of the skeletal remains of Josef Mengele by DNA analysis. *Forensic Sci. Int.* 56:65–76
 73. Kohn M, Wayne R. 1997. Facts from feces revisited. *Trends Ecol. Evol.* 6:223–27
 74. Kohn MH, York EC, Kamradt DA, Haught G, Sauvajot RM, Wayne RK. 1999. Estimating population size by genotyping faeces. *Proc. R. Soc. London Ser. B* 266:657–63
 75. Krajewski C, Driskell A, Baverstock P, Braun M. 1992. Phylogenetic relationships of the thylacine (Mammalia, Thylacinidae) among dasyuroid marsupials—evidence from *cytochrome b* DNA sequences. *Proc. R. Soc. London Ser. B* 50:19–27
 76. Krings M, Stone A, Schmitz RW, Krainitzski H, Stoneking M, Pääbo S. 1997. Neanderthal DNA sequences and the origin of modern humans. *Cell* 90:19–30
 77. Krings M, Geisert H, Schmitz RW, Krainitzski H, Pääbo S. 1999. DNA sequence of the mitochondrial hypervariable region II from the Neanderthal type specimen. *Proc. Natl. Acad. Sci. USA* 96:5581–85
 78. Kwok S, Higuchi R. 1989. Avoiding false positives with PCR. *Nature* 339:237–38
 79. Lalueza C, Pérez-Pérez A, Prats E, Cornudella L, Turbón D. 1997. Lack of founding Amerindian mitochondrial DNA lineages in extinct aborigines from Tierra del Fuego-Patagonia. *Hum. Mol. Genet.* 6:41–46
 80. Lawlor D, Dickel C, Hauswirth W, Parham P. 1991. Ancient HLA genes from 7500-year-old archaeological remains. *Nature* 349:785–88
 81. Lehman N. 1998. Conservation biology: genes are not enough. *Curr. Biol.* 8:722–24
 82. Lindahl T. 1993. Instability and decay of the primary structure of DNA. *Nature* 362:709–15
 83. Lindahl T. 1993. Recovery of antediluvian DNA. *Nature* 365:700
 84. Lindahl T. 1997. Facts and artifacts of ancient DNA. *Cell* 90:1–3

85. Logan GA, Boon JJ, Eglinton G. 1993. Structural biopolymer preservation in Miocene leaf fossils from the Clarkia site, northern Idaho. *Proc. Natl. Acad. Sci. USA* 90:2246–50
86. Lowenstein JM. 1985. Radioimmune assay of mammoth tissue. *Acta Zool. Fennica* 170:233–35
87. Matisoo-Smith E, Allen JS, Ladefoged TN, Roberts RM, Lambert DM. 1997. Ancient DNA from Polynesian rats: extraction, amplification and sequence from single small bones. *Electrophoresis* 18:1534–37
88. Matuschka F-R, Ohlenbusch A, Eiffert H, Richter D, Spielman A. 1996. Characteristics of Lyme disease spirochetes in archived European ticks. *J. Infect. Dis.* 174:424–26
89. McKusick V. 1991. The defect in Marfan syndrome. *Nature* 352:279–81
90. Merriwether DA, Rothhammer F, Ferrell RE. 1994. Genetic variation in the new world: ancient teeth, bone, and tissue as sources of DNA. *Experientia* 50:592–601
91. Monsalve M, Cardenas F, Guhl F, Delaney AD, Devine DV. 1996. Phylogenetic analysis of mtDNA lineages in South American mummies. *Ann. Hum. Genet.* 60:293–303
92. Morin P, Woodruff DS. 1996. Noninvasive genotyping for vertebrate conservation. In *Molecular Genetic Approaches in Conservation*, ed. TB Smith, RK Wayne, pp. 298–313. New York: Oxford Univ. Press
93. Morin PA, Moore JJ, Woodruff DS. 1992. Identification of chimpanzee subspecies with DNA from hair and allele-specific probes. *Proc. R. Soc. London Ser. B* 249: 293–97
94. Mullis KB, Faloona FA. 1987. Specific synthesis of DNA in vitro via a polymerase catalyzed chain reaction. *Meth. Enzymol.* 155:335–50
95. Mundy NI, Winchell CS, Burr T, Woodruff DS. 1997. Microsatellite variation and microevolution in the critically endangered San Clemente Island loggerhead shrike (*Lanius ludovicianus mearnsi*). *Proc. R. Soc. London Ser. B* 264:869–075
96. Noro M, Masuda R, Dubrovo IA, Yoshida MC, Kato M. 1998. Molecular phylogenetic inference of the woolly mammoth *Mammuthus primigenius*, based on complete sequences of mitochondrial cytochrome *b* and 12S ribosomal RNA genes. *J. Mol. Evol.* 46:314–26
97. O'Rourke DH, Carlyle SW, Parr RL. 1996. Ancient DNA: methods, progress and perspectives. *Am. J. Hum. Biol.* 8:557–71
98. Ozawa T, Hayashi S, Mikhelson VM. 1997. Phylogenetic position of mammoth and Steller's sea cow within Tethytheria demonstrated by mitochondrial DNA sequences. *J. Mol. Evol.* 44:406–13
99. Pääbo S. 1984. Über den Nachweis von DNA in Altägyptischen Mumien. *Das Altertum* 30:213–18
100. Pääbo S. 1985. Molecular cloning of ancient Egyptian mummy DNA. *Nature* 314:644–45
101. Pääbo S. 1989. Ancient DNA: extraction, characterization, molecular cloning and enzymatic amplification. *Proc. Natl. Acad. Sci. USA* 86:1939–43
102. Pääbo S. 1993. Ancient DNA. *Sci. Am.* 269:86–92
103. Pääbo S, Higuchi R, Wilson A. 1989. Ancient DNA and the polymerase chain reaction. *J. Biol. Chem.* 264:9709–12
104. Pääbo S, Irwin D, Wilson A. 1990. DNA damage promotes jumping between templates during enzymatic amplification. *J. Biol. Chem.* 265:4718–21
105. Pääbo S, Wilson A. 1991. Miocene DNA sequences: a dream come true? *Curr. Biol.* 1:45–46
106. Palumbi SR, Cipriano F. 1998. Species identification using genetic tools: the value of nuclear and mitochondrial gene sequences in whale conservation. *J. Hered.* 89:459–64
107. Parr R, Carlyle S, O'Rourke D. 1996. Ancient DNA analysis of Fremont Amerindians of the Great Salt Lake wetlands. *Am. J. Phys. Anthropol.* 99:507–18

108. Persing D, Tedford III S, Rys PN, Dodge DE, White TJ, et al. 1990. Detection of *Borrelia burgdorferi* in DNA in museum specimens of *Ixodes dammini* ticks. *Science* 249:1420–23
109. Poinar G. 1994. The range of life in amber: significance and implications in DNA studies. *Experientia* 50:536–42
110. Poinar H, Cano R, Poinar G. 1993. DNA from an extinct plant. *Nature* 363:677
111. Poinar HN, Hofreiter M, Spaulding WG, Martin PS, Stankiewicz BA, et al. 1998. Molecular coproscopy: dung and diet of the extinct ground sloth *Nothrotheriops shastensis*. *Science* 281:402–6
112. Poinar HN, Höss M, Bada JL, Pääbo S. 1996. Amino acid racemization and the preservation of ancient DNA. *Science* 272:864–66
113. Rafi A, Spigelman M, Stanford J, Lemma E, Donoghue H, et al. 1994. *Mycobacterium leprae* DNA from ancient bone detected by PCR. *Lancet* 343:1360–61
114. Reed J, Tollit D, Thompson P, Amos W. 1997. Molecular scatology: the use of molecular genetic analysis to assign species, sex and individual identity to seal faeces. *Mol. Ecol.* 6:225–34
115. Reid AH, Fanning TG, Hultin JV, Taubenberger JK. 1999. Origin and evolution of the 1918 “Spanish” influenza virus hemagglutinin gene. *Proc. Natl. Acad. Sci. USA* 96:1651–56
116. Roelke-Parker ME, Munson L, Packer C, Kock R, Cleaveland S, et al. 1996. A canine distemper virus epidemic in Serengeti lions *Panthera leo*. *Nature* 379:441–45
117. Rogers AR, Harpending H. 1992. Population growth makes waves in the distribution of pairwise genetic differences. *Mol. Biol. Evol.* 9:552–69
118. Rollo F, Venanzi F, Amici A. 1991. Nucleic acids in mummified plant seeds: biochemistry and molecular genetics of pre-Columbian maize. *Genet. Res.* 58:193–201
119. Roy M, Girman D, Wayne R. 1994. The use of museum specimens to reconstruct the genetic variability and relationships of extinct populations. *Experientia* 50:551–63
120. Roy MS, Geffen E, Smith D, Wayne RK. 1996. Molecular genetics of pre-1940 red wolves. *Conserv. Biol.* 10:1413–24
121. Salo W, Aufderheide A, Buikstra J, Holcomb T. 1994. Identification of *Mycobacterium tuberculosis* DNA in a pre-Columbian Peruvian mummy. *Proc. Natl. Acad. Sci. USA* 91:2091–94
122. Salvo J, Allison M, Rogan P. 1989. Molecular genetics of pre-Columbian South American mummies. *Am. J. Phys. Anthropol.* 78:295
123. Shoshani J, Walz DA, Goodman M, Lowenstein JM, Prychodko W. 1985. Protein and anatomical evidence of the phylogenetic position of *Mammuthus primigenius* within the Elephantinae. *Acta Zool. Fennica* 170:237–40
124. Sidow A, Wilson A, Pääbo S. 1991. Bacterial DNA in *Clarkia* fossils. *Philos. Trans. R. Soc. London Ser. B* 333:429–33
125. Smith F, Betancourt J, Brown JH. 1995. Evolution of body size in the woodrat over the past 25,000 years of climate change. *Science* 270:2012–14
126. Soltis P, Soltis D, Smiley C. 1992. An rbcL sequence from a Miocene *Taxodium* (bald cypress). *Proc. Natl. Acad. Sci. USA* 89:499–501
127. Soltis PS. 1995. Fossil DNA: its potential for biosystematics. In *Experimental and Molecular Approaches to Plant Biosystematics*, ed. PC Hoch, AG Stephenson, pp. 1–13. St. Louis: Missouri Bot. Gard.
128. Soltis PS, Soltis DE. 1993. Ancient DNA: prospects and limitations. *New Zealand J. Bot.* 31:203–9
129. Stankiewicz BA, Poinar HN, Briggs DEG, Evershed RP, Poinar GO Jr. 1998. Chemical preservation of plants and insects in natural resins. *Proc. R. Soc. London Ser. B* 265:641–47

130. Steadman DW. 1995. Prehistoric extinctions of Pacific island birds: biodiversity meets zooarchaeology. *Science* 267: 1123–31
131. Stone AC, Milner GR, Pääbo S, Stoneking M. 1996. Sex determination of ancient human skeletons using DNA. *Am. J. Phys. Anthropol.* 99:231–38
132. Stone AC, Stoneking M. 1993. Ancient DNA from a pre-Columbian Amerindian population. *Am. J. Phys. Anthropol.* 92: 463–71
133. Stoneking M. 1995. Ancient DNA: How do you know when you have it and what can you do with it? *Am. J. Hum. Genet.* 57:1259–62
134. Swofford DL, Olsen GJ, Waddell PJ, Hillis DM. 1996. Phylogenetic inference. In *Molecular Systematics*, ed. DM Hillis, C Moritz, BK Mable, pp. 407–514. Sunderland, MA: Sinauer. 2nd ed.
135. Sykes B. 1991. Ancient DNA: the past comes alive. *Nature* 352:381–82
136. Sykes B. 1993. Ancient DNA: less cause for grave concern. *Nature* 366:513
137. Taberlet P, Bouvet J. 1992. Bear conservation genetics. *Nature* 358:197
138. Taberlet P, Fumagalli L. 1996. Owl pellets as a source of DNA for genetic studies of small mammals. *Mol. Ecol.* 5:301–5
139. Taylor G, Crossey M, Saldanha J, Waldron T. 1996. DNA from *Mycobacterium tuberculosis* identified in mediaeval human skeletal remains using polymerase chain reaction. *J. Archaeol. Sci.* 23:789–98
140. Taylor JW, Swann EC. 1994. DNA from herbarium specimens. In *Ancient DNA*, ed. B Hermann, S Hummel, pp. 166–81. New York: Springer-Verlag
141. Templeton AR, Routman E, Phillips CA. 1995. Separating population structure from population history: a cladistic analysis of the geographical distribution of mitochondrial DNA haplotypes in the Tiger salamander, *Ambystoma tigrinum*. *Genetics* 140:619–33
142. Thomas R, Schaffner W, Wilson A, Pääbo S. 1989. DNA phylogeny of the extinct marsupial wolf. *Nature* 340:465–67
143. Thomas WK, Pääbo S, Villablanca FX, Wilson AC. 1990. Spatial and temporal continuity of kangaroo rat populations shown by sequencing mitochondrial DNA from museum specimens. *J. Mol. Evol.* 31:101–12
144. Vane-Wright RI. 1996. Systematics and the conservation of biological diversity. *Ann. Missouri Bot. Gard.* 83:47–57
145. Walden KKO, Robertson HM. 1997. Ancient DNA from amber fossil bees? *Mol. Biol. Evol.* 14:1075–77
146. Wang H-L, Yan Z-Y, Jin D-Y. 1997. Reanalysis of published DNA sequence amplified from Cretaceous dinosaur egg fossil. *Mol. Biol. Evol.* 14:589–91
147. Wayne R, Jenks S. 1991. Mitochondrial DNA analysis implies extensive hybridization of the endangered red wolf *Canis rufus*. *Nature* 351:565–68
148. Weiss KM, Buchanan AV, Daniel C, Stoneking M. 1994. Optimizing utilization of DNA from rare of archival anthropological samples. *Hum. Biol.* 66:789–804
149. Williams S, Longmire J, Beck L. 1990. Human DNA recovery from ancient bone. *Am. J. Phys. Anthropol.* 81:318
150. Woodward SR, Weyand NJ, Bunnell M. 1994. DNA sequence from Cretaceous period bone fragments. *Science* 266: 1229–32
151. Wright S. 1978. *Evolution and the Genetics of Populations*. Vol. 4: *Variability Within and Among Natural Populations*. Chicago: Univ. Chicago Press
152. Yang H, Golenberg E, Shoshani J. 1996. Phylogenetic resolution within the Elephantidae using fossil DNA sequence from the American mastodon (*Mammot americanum*) as an outgroup. *Proc. Natl. Acad. Sci. USA* 93:1190–94
153. Young DL, Huyen Y, Allard MW. 1995. Testing the validity of the cytochrome *b*

sequence from Cretaceous period bone fragments as dinosaur DNA. *Cladistics* 11:199–209

154. Yousten AA, Rippere KE. 1997. DNA similarity analysis of a putative ancient bacterial isolate obtained from amber. *FEMS Microbiol. Lett.* 152:345–47
155. Zischler H, Höss M, Handt O, von Haeseler A, van der Kuyl AC, et al. 1995. Detecting dinosaur DNA. *Science* 268:1192
156. Zischler H, Geisert H, von Haeseler A, Pääbo S. 1995. A nuclear ‘fossil’ of the mitochondrial D-loop and the origin of modern humans. *Nature* 378:489–92



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